When Measurements Become Targets: The Strategic Manipulation Challenge for Internet Observatories

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ABSTRACT

Internet observatories increasingly influence reputation, compliance assessments, and business relationships across the Internet ecosystem. Yet many of these observatories depend on measurement infrastructures that are public, long-lived, and therefore potentially recognizable to the networks they evaluate. As favorable measurement outcomes become increasingly valuable, operators may have incentives to optimize specifically for measurement platforms while behaving differently toward ordinary users and traffic. We refer to this phenomenon as *measurement awareness* and argue that it represents an emerging challenge for Internet measurement.

Using Route Origin Validation (ROV) as a case study, we compare observations obtained using publicly known RPKI-invalid targets with those obtained using functionally equivalent but undisclosed shadow targets. We identify networks exhibiting behavior consistent with measurement awareness, rejecting well-known invalid prefixes while continuing to accept equivalent shadow prefixes. Our findings suggest that Internet observatories may not always capture the properties they intend to measure, but rather the behavior networks choose to expose to the measurement infrastructure.

1 INTRODUCTION

Internet measurement plays a central role in understanding the operational state of the network [8]. Researchers, operators, regulators, and policymakers rely on measurement studies to evaluate properties ranging from performance and reliability [11, 19] to routing security [13, 16, 17] and censorship [14, 23, 29]. Many of these studies rely on long-lived Internet observatories and shared measurement infrastructures that provide continuous visibility into network behavior.

Although side-channel techniques [10, 22] can sometimes extend visibility beyond dedicated measurement infrastructures, they often provide limited experimental control and may disappear as implementations evolve [17]. In contrast, many measurement studies require stable platforms that remain comparable over long periods of time, making continuity and reproducibility valuable properties of Internet observatories.

The influence of these observatories increasingly extends beyond academic research. Measurements are routinely incorporated into security assessments, compliance initiatives, operational decisions, and commercial relationships. Consequently, favorable measurement outcomes can carry reputational and economic value for the networks being evaluated.

This evolution introduces an implicit assumption: that networks behave consistently regardless of who is observing them. In most studies, measurements are interpreted as reflecting the actual policies deployed by the network under observation rather than policies selectively exposed to the measurement process itself.

However, the characteristics that make Internet observatories attractive to researchers can also make them distinguishable to measured networks. Public documentation, repeated use of the same infrastructure, and predictable experimental methodologies may allow operators to recognize measurement activity and react differently to it than they would to ordinary traffic.

If such differentiation is possible, Internet observatories become vulnerable to strategic behavior. Rather than deploying a mechanism uniformly across their networks, operators may instead optimize specifically for the environments in which they know they are evaluated. This possibility raises a fundamental question for Internet measurement: are we measuring the Internet itself, or are we measuring the Internet's response to being measured?

In this paper, we investigate this question through the lens of routing security measurements and explore its implications for the design and interpretation of Internet observatories.

Our contributions are threefold:

- We introduce the notion of *measurement-aware networks*, in which operators selectively adapt their behavior in response to identifiable measurement activity.
- We propose a paired-target methodology using public and shadow measurement targets and develop a strategy for selecting representative RPKI-invalid prefixes.
- Using ROV as a case study, we identify networks exhibiting behavior consistent with measurement awareness.

2 MEASUREMENT-AWARE NETWORKS

Internet measurements implicitly rely on an *invariance assumption*: that networks behave identically regardless of

whether they are interacting with ordinary traffic or participating in a measurement experiment. Under this assumption, observations made by measurement platforms accurately reflect the policies deployed by the measured network.

We define a *measurement-aware network* as a network that recognizes measurement activity and selectively adapts its behavior to influence the resulting observations. Such adaptations may include selectively filtering routes, modifying forwarding behavior, prioritizing traffic, or altering reachability outcomes only for interactions associated with known measurement infrastructures.

Measurement-aware behavior introduces a new source of bias because networks may optimize for favorable measurement outcomes rather than uniformly deploy the mechanisms being measured. Consequently, Internet observatories may capture not only intrinsic network properties but also the behavior networks choose to expose to the measurement infrastructure.

2.1 Strategic Behavior Model

We consider a network operator seeking favorable outcomes in reputation systems, compliance frameworks, or public observatories.

Knowledge. The operator has access only to publicly available or externally observable information, including publicly documented measurement infrastructures and published methodologies.

Capabilities. The operator can distinguish measurement traffic from ordinary traffic and apply different policies to the two classes, for example by selectively filtering routes, prioritizing traffic, or altering reachability outcomes.

Objective. The operator aims to obtain favorable measurement results without uniformly deploying the corresponding mechanisms throughout its network, selectively applying the desired behavior only to traffic identified as belonging to a measurement campaign.

This model requires neither insider knowledge nor specialized capabilities. As illustrated in Figure 1, a network operator can implement measurement-aware policies using standard filtering and policy mechanisms already supported by commodity routers.

2.2 Identifiable Measurement Setups

Internet measurement infrastructures often expose information that allows measured networks to recognize measurement activity. Depending on the methodology, this information may include publicly documented probe infrastructures such as RIPE Atlas and CAIDA Ark [5, 25], observation vantage points such as BGP collectors [26, 30], or experimental infrastructures such as the PEERING testbed [27], where measurement resources are intentionally disclosed to support reproducibility, facilitate independent validation of results, and provide visibility into the deployment status of operational and security mechanisms.

Previous work has shown that adversaries can exploit knowledge of monitoring infrastructure to evade detection from known BGP collectors [3, 4], demonstrating that observability itself can become a target for strategic behavior.

Identification does not necessarily require explicit disclosure, however. In our own analysis of ROV measurement platforms, we were able to infer previously undisclosed experimental prefixes using only publicly available methodology descriptions and observations from public BGP collectors. This suggests that measurement infrastructures may become identifiable even when specific measurement targets are not intentionally published.

More broadly, previous work has demonstrated that network activity can often be identified from externally observable characteristics even when explicit identifiers are absent. For example, OpenVPN connections can be accurately identified through passive fingerprinting and active probing despite attempts at obfuscation [31]. In comparison, Internet measurement infrastructures frequently rely on publicly documented resources, long-lived deployments, and repeated experimental methodologies, all of which may contribute to their recognizability.

2.3 Detection Methodology

Detecting measurement-aware behavior is inherently challenging because researchers typically observe only measurement outcomes and cannot directly determine whether a network behaved differently in response to the measurement process itself.

To address this challenge, we propose a paired-target methodology based on two functionally equivalent measurement targets. The first is a *public target*, which is publicly associated with the measurement infrastructure and therefore potentially recognizable to the measured network. The second is a *shadow target*, which provides an equivalent measurement function but is not publicly associated with the measurement campaign.

Measurements are performed simultaneously toward both targets under identical experimental conditions. Under uniform policy deployment, a network should exhibit equivalent behavior toward public and shadow targets. Systematic discrepancies between the two observations may therefore indicate that the network is adapting its behavior based on recognition of the measurement infrastructure rather than applying a consistent operational policy.

The specific realization of public and shadow targets depends on the measurement domain. In the remainder of

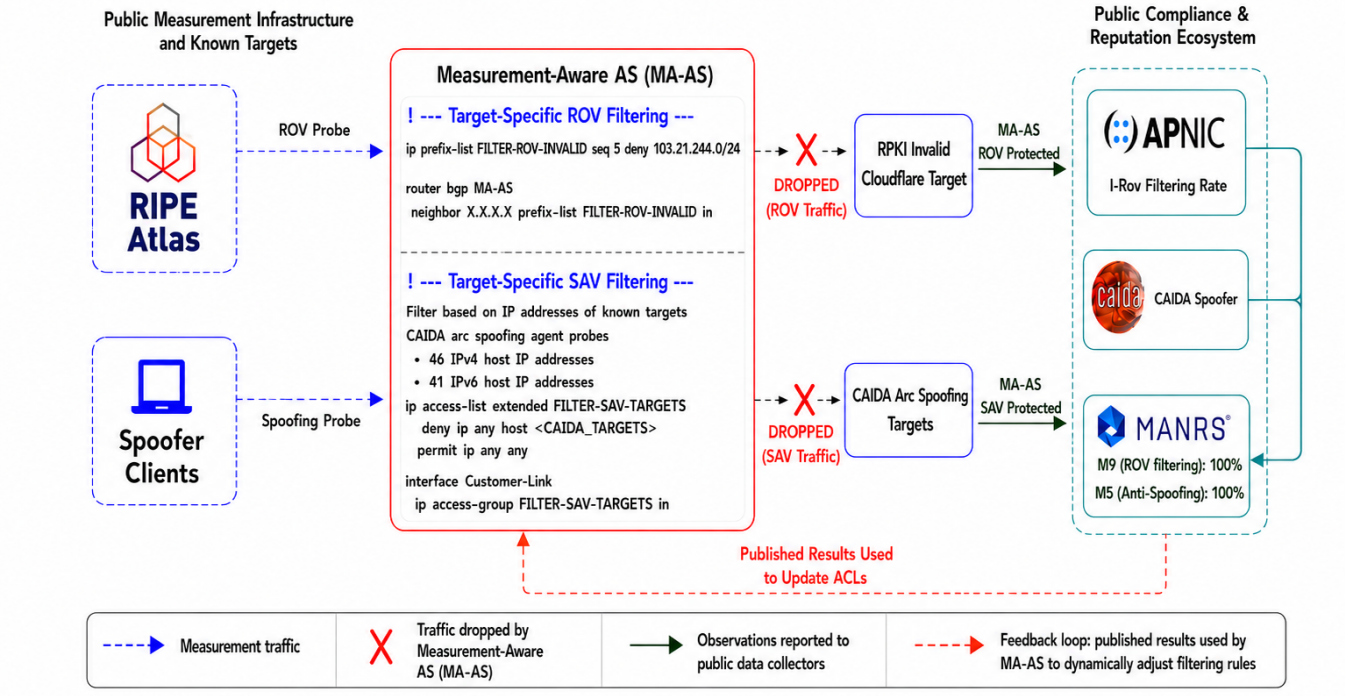


Figure 1: Conceptual overview of a measurement-aware AS (MA-AS). The MA-AS identifies traffic associated with publicly known measurement infrastructure and selectively applies filtering policies to influence measurement outcomes in its favor. The resulting observations are then incorporated into compliance and reputation frameworks, creating incentives for strategic behavior and a feedback loop in which published measurements influence future network policies. Configuration snippets are illustrative examples of mechanisms commonly available in commodity routers and network operating systems.

this paper, we instantiate this methodology using ROV measurements based on publicly known and undisclosed RPKI-invalid prefixes.

3 ROV CASE STUDY

ROV [20] uses Route Origin Authorizations (ROAs) [28] published through the Resource Public Key Infrastructure (RPKI) [15] to validate the origin AS of BGP announcements. Routes are classified as *Valid*, *Invalid*, or *Unknown*, and operators may reject or de-preference routes classified as invalid.

Several measurement platforms infer ROV deployment by comparing the reachability of valid and invalid prefixes from distributed vantage points. Such approaches typically rely on a small number of publicly known invalid prefixes and long-lived measurement infrastructures [2, 13, 16]. As a result, these measurement targets may themselves become recognizable to the networks being evaluated.

Recent systems such as RScope [17] have substantially improved the accuracy of ROV measurements by addressing limitations of earlier methodologies. However, even these approaches continue to depend on a limited set of invalid test prefixes and measurement targets, which are often reused across studies and over extended periods of time, increasing their visibility to measured networks.

Figure 3 illustrates this effect using publicly observable data from RoVista. While the platform relies on RPKI-invalid prefixes observed in the wild rather than dedicated test prefixes, the set of active invalid targets has progressively narrowed, with only five prefixes remaining active since January 2026. The resulting concentration on a small and stable set of measurement targets increases their visibility to measurement-aware networks.

ROV therefore provides a natural case study for measurement awareness. A network seeking favorable ROV scores could selectively filter publicly known invalid prefixes while

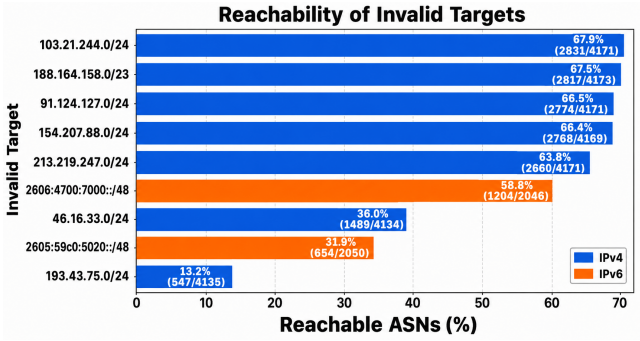


Figure 2: Reachability of candidate RPKI-invalid measurement targets from unique source ASes. Reachability varies substantially across prefixes, ranging from 13% to 67% of source ASes. The large differences in visibility highlight the importance of target selection in ROV measurements and motivate the use of carefully matched public and shadow targets.

continuing to accept equivalent invalid routes that are not publicly associated with measurement campaigns. Existing observatories would interpret such behavior as evidence of ROV deployment despite the absence of uniform filtering policies.

To investigate this possibility, we compare observations obtained using publicly known invalid prefixes with observations obtained using equivalent shadow targets. Figure 2 shows the fraction of source ASes that can reach each candidate invalid prefix. Reachability varies substantially across targets, ranging from approximately 13% to 67% of source ASes. This variability highlights an additional challenge for ROV measurements: different invalid prefixes may expose different portions of the Internet, making measurement outcomes sensitive to the choice of target itself.

3.1 Selecting Public and Shadow Targets

Applying the paired-target methodology to ROV requires public and shadow targets that are comparable in their routing characteristics and global visibility.

As our public targets, we use the publicly known RPKI-invalid prefixes employed by the *Is BGP Safe Yet?* project and subsequently adopted by APNIC's ROV observatory and MANRS compliance framework. The prefixes marked with † in Table 1 correspond to these public targets, while the remaining high-reachability prefixes serve as shadow targets.

To construct suitable shadow targets, we seek RPKI-invalid prefixes that are not publicly associated with ROV measurement efforts, but are propagated through the same Cloudflare anycast infrastructure and exhibit reachability characteristics comparable to those of the public targets. Using prefixes announced through the same anycast platform minimizes differences in routing behavior and global visibility, reducing the likelihood that observed discrepancies arise from infrastructure artifacts rather than measurement awareness. To further improve comparability, we exclude prefixes whose observed reachability could be explained by valid or unknown covering prefixes rather than the invalid prefix itself.

While the public targets are originated directly by AS13335 and the shadow targets originate from third-party ASes (AS209242 and AS63888), both target classes ultimately rely on Cloudflare's anycast infrastructure for route propagation. From a control-plane perspective, AS13335 originates the public targets and appears as the immediate upstream AS for all observed announcements of the shadow targets, resulting in highly similar propagation paths and visibility characteristics. From a data-plane perspective, traceroute measurements toward both public and shadow targets traverse Cloudflare's network and do not cross additional external transit ASes before reaching their final destination within the origin network. Consequently, the two target classes differ primarily in their public association with ROV measurement activities rather than in their routing properties or global visibility.

To identify suitable candidates, we collect RPKI-invalid prefixes observed across multiple BGP collectors using bgproutes.io [1] and rank them according to a *Prefix Reachability Score* (PRS):

$$PRS = w_1 \cdot V + w_2 \cdot G + w_3 \cdot C + w_4 \cdot P \quad (1)$$

where V denotes prefix visibility, G the geographic diversity of observing collectors, C the diversity of collector ASes, and P the diversity of observed AS paths. All components are normalized to the range $[0, 100]$ and weighted equally.

Candidate prefixes are grouped into high-, medium-, and low-reachability categories according to their PRS (Table 1). Our measurement-aware analysis focuses on the high-reachability category, which provides both sufficient visibility and a comparable set of public and shadow targets.

3.2 Detecting Measurement-Aware ROV Deployment

We perform traceroute measurements using RIPE Atlas probes toward both target classes from 11,384 probes distributed across 4,171 ASes over a period of 14 days. Traceroute interfaces are mapped to ASes using Prefix2Org [12] datasets and supplemented with PeeringDB and PCH [21, 24] information to account for IXP traversals.

By comparing routing behavior toward both target classes, we classify an AS as exhibiting behavior consistent with measurement awareness if:

(1) publicly known invalid targets are unreachable from probes within the AS;

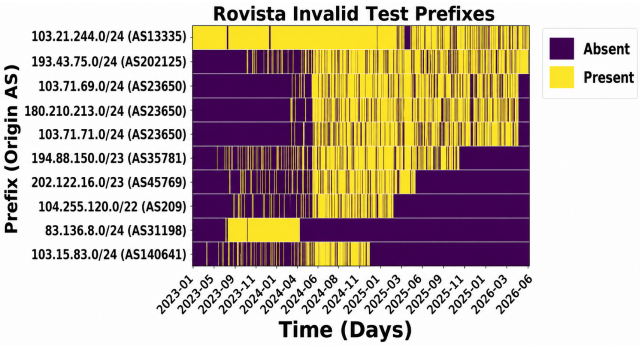


Figure 3: RPKI-invalid test prefixes used by RoVista between January 2023 and June 2026. The number of active invalid targets decreases over time, with only five prefixes remaining active since January 2026. The limited and stable set of measurement targets increases their visibility to measured networks and may facilitate measurement-aware behavior.

Table 1: RPKI-invalid prefixes considered in this study, grouped by visibility score. † denotes publicly known measurement targets used by existing ROV observatories.

Prefix	Origin AS	Score
High-reachability targets		
188.164.158.0/23	209242	67.58
154.207.88.0/24	63888	66.77
103.21.244.0/24†	13335	66.52
213.219.247.0/24	209242	63.46
91.124.127.0/24	209242	62.72
2606:4700:7000::/48†	13335	56.47
Medium-reachability targets		
46.16.33.0/24	208890	38
2605:59c0:5020::/48	45700	35.62
Low-reachability targets (RoVista)		
193.43.75.0/24	202125	22.21

(2) equivalent shadow invalid targets remain reachable from the same probes; and
(3) the AS path observed toward the shadow invalid target is consistent with that of the corresponding valid control announcement used by *Is BGP Safe Yet?*.

To attribute the observed behavior to the measured AS rather than to downstream transit providers, we further restrict our analysis to cases where traceroutes toward the valid control announcement terminate within Cloudflare directly after the measured AS, while traceroutes toward the shadow invalid targets follow the path:

$$\textit{Measured AS} \longrightarrow \textit{AS13335} \longrightarrow \textit{Shadow Origin AS}$$

where the shadow origin AS corresponds to either AS209242 or AS63888. Under these conditions, discrepancies between public and shadow observations necessarily arise before traffic enters the shadow origin network and therefore cannot be attributed to filtering performed by the shadow origin AS or intermediate downstream transit providers.

An AS that rejects publicly known invalid announcements while continuing to accept equivalent shadow invalids exhibits behavior that is difficult to reconcile with ordinary ROV policies and is instead consistent with measurement-aware deployment.

While alternative explanations such as partial deployment, router-level policy differences, or configuration inconsistencies cannot be fully excluded, they do not readily explain why discrepancies consistently align with publicly known observatory targets while equivalent shadow targets remain reachable.

Because routing policies may be implemented at individual routers or points of presence rather than uniformly across an AS, different probes within the same network may observe different outcomes. We therefore distinguish between *strictly measurement-aware* ASes, where all observing probes exhibit the same behavior, and *partially measurement-aware* ASes, where observations differ across vantage points.

Our measurements identify 13 strictly measurement-aware ASes and an additional 2 partially measurement-aware ASes for which the behavior was observed only for IPv4 announcements.

To evaluate whether these observations could be explained by transient routing dynamics or short-lived configuration artifacts, we performed an additional longitudinal analysis over a period of one month for all identified measurement-aware ASes. In all cases, the observed discrepancies between public and shadow targets remained stable throughout the observation period, suggesting that the behavior reflects persistent routing policies rather than temporary routing events or time-specific artifacts.

These networks span all five Regional Internet Registry (RIR) regions and exhibit customer cone sizes ranging from 0 to 161 downstream ASes according to CAIDA AS Rank [6], suggesting that measurement-aware behavior is not confined to any particular geography or operator class.

Although the absolute number of identified ASes is modest, our objective is not to estimate prevalence but rather to demonstrate existence. The observation of even a small number of networks exhibiting behavior inconsistent with uniform ROV deployment is sufficient to challenge the assumption that observatory measurements are necessarily measurement-independent.

Despite accepting shadow invalid announcements, the identified measurement-aware networks receive remarkably high ROV deployment scores from existing observatories.

Across our dataset, APNIC reports a median ROV score of 90.3%, while RoVista reports a median score of 100%. The fact that networks exhibiting measurement-aware behavior can nevertheless appear as near-perfect RPKI-ROV adopters exposes a fundamental challenge for current measurement methodologies.

4 DISCUSSION

4.1 Beyond ROV

Although our evaluation focuses on ROV, the underlying vulnerability is not specific to routing security measurements. We do not experimentally evaluate the following domains; rather, they illustrate how measurement-aware behavior could affect a broader class of Internet observatories.

Source Address Validation (SAV) provides one example. Projects such as CAIDA's Spoofer [7] rely on a publicly known set of monitoring destinations to evaluate the Internet's susceptibility to IP spoofing, and these measurements are subsequently incorporated into initiatives such as the MANRS Observatory [9, 18]. An operator seeking favorable SAV scores could selectively enforce source address validation only for traffic directed toward known measurement infrastructure while continuing to permit spoofed traffic elsewhere. Such a network would appear compliant to the observatory despite lacking comprehensive SAV deployment.

Censorship measurements face a similar challenge due to their reliance on stable vantage points [29]. A censoring network aware of the addresses used by measurement platforms could selectively permit traffic originating from known probes while continuing to restrict access for ordinary users. Measurement platforms would observe successful connectivity, whereas users within the network would continue to experience censorship.

Performance measurements may also be vulnerable. Traffic originating from known measurement infrastructure could be prioritized through dedicated QoS policies, resulting in latency, packet loss, and throughput measurements that do not accurately reflect the experience of ordinary users.

Taken together, these examples suggest that measurement awareness may represent a general challenge for Internet observatories whenever measurement infrastructures become stable, identifiable, and associated with incentives for favorable outcomes.

4.2 Measurement-Aware Detection and Mitigation

Detecting measurement-aware behavior is challenging because observatories typically observe only measurement outcomes rather than the decision processes that produced them. Consequently, measurement-aware networks cannot be identified through direct inspection and must instead be inferred from inconsistencies in observed behavior.

Our approach treats the Measurement-Aware AS (MA-AS) illustrated in Figure 1 as a black box whose behavior is inferred through differential measurements. Rather than attempting to determine how a network recognizes measurement traffic, we focus on identifying discrepancies that are difficult to explain under uniform policy deployment.

One possible mitigation strategy is to introduce diversity on both the source and destination sides of the measurement infrastructure. On the destination side, publicly known measurement targets can be complemented with undisclosed shadow targets that provide equivalent measurement functionality while remaining unknown to the measured network. On the source side, measurements originating from publicly recognizable infrastructures such as RIPE Atlas probes could be complemented with measurements originating from shadow probes that are not publicly associated with measurement campaigns.

These two forms of diversity provide complementary signals regarding measurement-aware behavior. Source-side discrepancies (Δ_{source}) capture differences between observations obtained from public and shadow probes toward identical destinations, while destination-side discrepancies (Δ_{target}) capture differences between observations obtained toward public and shadow targets using identical sources.

Under ordinary operation, a network applying a uniform policy should produce comparable observations regardless of the visibility of the probe or destination involved in the measurement. Significant differences in either signal therefore suggest that the measured network may be adapting its behavior in response to identifiable measurement infrastructure rather than applying a consistent operational policy.

While such approaches cannot eliminate measurement awareness entirely, they substantially increase the cost of strategic manipulation and reduce the incentives for selectively optimizing behavior toward Internet observatories. Rather than assuming that measurements directly reflect network properties, future observatories may need to explicitly account for the possibility that measured networks adapt their behavior in response to being observed.

4.3 Ethical Considerations

Our measurements rely exclusively on standard RIPE Atlas traceroutes and publicly observable BGP information. The experiments do not collect user data, inspect payload contents, or interfere with production traffic. The shadow targets used in our study are functionally equivalent to existing ROV measurement targets and differ primarily in their public visibility rather than in the routing behavior they induce. Finally, our

objective is to characterize potential sources of measurement bias rather than to publicly attribute intent or behavior to individual network operators.

5 CONCLUSION

Internet observatories implicitly assume that measured networks behave consistently toward measurement infrastructure and ordinary users. In this paper, we introduced the notion of *measurement-aware networks*, in which operators recognize measurement activity and selectively adapt their behavior to influence observed outcomes. Using ROV as a case study, we proposed a paired-target methodology based on public and shadow measurement targets and identified networks exhibiting behavior consistent with measurement awareness. Our findings suggest that Internet observatories may increasingly measure the behavior networks choose to expose to measurement platforms rather than the underlying properties they intend to characterize. Future measurement studies should therefore consider strategic behavior by measured networks as an additional source of measurement bias.

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